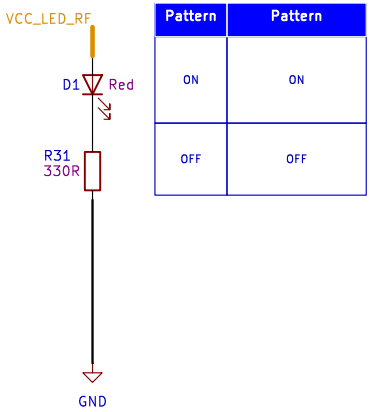
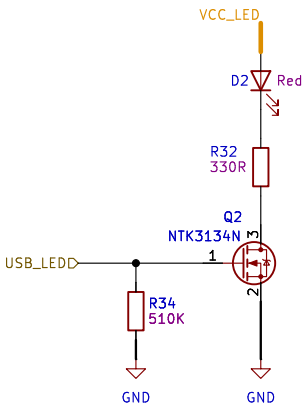
[7] User IO



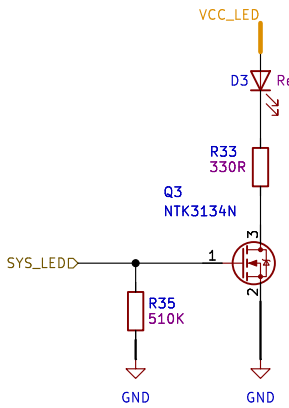
Pattern	Pattern
ON	ON
OFF	OFF

NFC activity LED



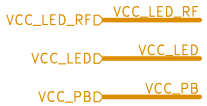
Pattern	State
ON	Host has enumerated and mounted/opened the device (USB configured, crypto-token session open where applicable). Do not unplug.
OFF	Not on USB power, not yet enumerated, or session cleanly closed. Safe to unplug.
Blinking	

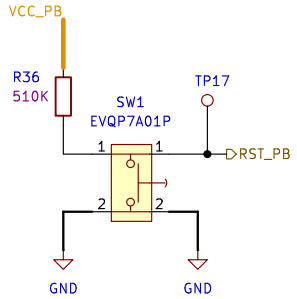
USB Status LED



Pattern	State
500 ms ON, 1500 ms OFF, repeating	Bootloader / DFU active
3× (50 ms ON, 200 ms OFF), one-shot	Application start
5× (50 ms ON, 50 ms OFF), one-shot per event	NFC event while powered
200 ms ON, 200 ms OFF, repeating	Crypto operation in progress
20 ms ON every 2000 ms	BLE advertising
2x 20 ms ON every 2000 ms	BLE connected
1000 ms ON, 1000 ms OFF, repeating	USB charging in progress
Solid ON	USB charging complete
100 ms ON, 100 ms OFF, continuous	Error
OFF	Idle

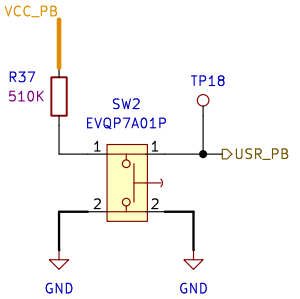
System Status LED





Action	Result
Push	Reset the device

Reset Button



Action	Result
Maintaid Pushed while booting	Stay in bootloader for DFU operation
Short push	Action 1
Short push	Action 1
Short push	Action 1

Boot/User Button

Comments:	Company:		Variant: DRAFT	Git Hash: d5f0558
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: User IO	File Name: Leds.kicad_sch	Designer: Antonio Morales	Date: 2026-09-08	Revision:
Sheet Path: /Project Architecture/Leds/		Reviewer:	Size: **A4**	Sheet: **7** of **10**